Abstract

Bacteriophage therapy is emerging as a promising alternative to antibiotics against multi drug resistant bacteria, but its clinical use depends on correctly matching a phage to a bacterial host it can infect. Existing computational host prediction tools return point predictions or uncalibrated scalar scores, with no finite sample guarantee that the true host is among the candidates returned.

This report develops a distribution free conformal prediction framework for phage host classification that closes this gap. We build on the Conformal Set Aggregation (CSA) approach and replaced its convex calibration envelopes, which may become uninformatively large when the class conditional score clusters are well separated. We used three star shaped, non convex alternatives (Radial, Strip, and Collapsed 1D) which are suited to the disjoint geometry of classification tasks. For each geometry we derived a closed form calibration rule that replaces the iterative binary search used in standard CSA with a single sorting pass. We also introduce a class conditional calibration, which provides class conditional finite sample validity under the exchangeability assumptions within each host class, rather than a single marginal one. Finally, we use a two stage Gram type, to host cascade with a proven bound on compounded miscoverage.

provides class-conditional finite-sample validity under the exchangeability assumptions within each host class

We validated the framework on a real dataset of bacteriophage protein functional scores, and achieved reliable empirical coverage at the target confidence level on both a binary Gram type task and a multiclass host classification task spanning 54 candidate hosts. Gram type pre filtering also reduced average prediction set sizes by over 60% compared to an unfiltered baseline, and with no loss of coverage.

hello

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